MILITARY SPECIFICATION
CABLE ASSEMBLY, MAGNETIC SWEEP MARK 17 MOP L

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers electrodes, "S" and pigtail type cable for magnetic minesweeping.

1.2 Classification. Magnetic minesweeping cables shall be classified as follows:

Type Electrode - Noninsulated, single conductor buoyant.
Type "S" - Insulated - single condutor, buoyant.
Type Pigtail - Insulated, single conductor, nonbuoyant.

1.2.1 Standard lengths. Magnetic minsweeping cables shall be furnished in the standard lengths shown in table I as specified (see 6,1).

Table I - Standard legths

Cable length	Length Feet	Tolerance of length Plus(ft)	Minus(ft)
Electrode	150	2	1
"S"	450	5	0
Pigtail	4	0.2	0.2

1.2.2 Assembly. The magnetic minesweeping cable assembly shall consist of two .150-foot electrodes, one 450-foot "S" type cable, one pigtail type cable, and two insulating sleeves.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

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L-P-535 - Plastic Sheet (Sheeting and Film): Vinly Chloride Polymer and Vinyl Chloride-Vinyl Acetate Coplymer, Rigid.

UU-P-271- Paper, Kraft, Wrapping, Waterproofed.

MILITARY

MIL-B-16033, CL 1 - Bronze, Aluminum: Casting

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

DRAWINGS

MILITARY

S8101-1535866 - Details of 8-foot Diameter by 4-foot T by 4-Foot 8-Inch Shipping Reel for S1-S2 (Single Section).

NAVAL AIR SYSTEMS COMMAND

30003-506AS101 Cable-Assembly, Magnetic Sweep Mark 17 - Mof 1, General Assembly and "S" Cable Details

30003-506AS100 cable Assembly, Magnetic Sweep Mark 17 - Mod 1, Electrode Details

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from. the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. *The following documents* specification *to* the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS

American Association. of Railroads Rules.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren St., Chicago , Illinois 60605)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-G-23-69 - Oxygen and sunlight resistance

ASTM-B-230-71 - Hard-Drawn Aluminum Wire for Electrical Purpos~

ASTM-B-133-66 - Copper Rod, Bar, And Shapes

ASTM-B-75-72 - Seamless Copper Tube

(Application for copies should be addressed to the American Society for Testing and Materials,1916 Race Street, Philadelphia, Pa. 19103)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Commit-tee, 1 Park Avenue at 33rd St., New York , N.Y. 10015)

3. REQUIREMENTS

3.1 Material. Materials of construction shall be of the best quality and entirely suitable for the intended application.

3.1.1 Synthetic rubbers.

3.1.1.1 Insulation, Synthetic rubber compunds used in the insulation shall be butadiene-styrene copolymer (SBR), chlorosulphonated polyethylene having an insulation resistance K factor of not less than 1500 (Hypalon-20 or equal), or a combination thereof. They shall have high insulation resistance, flexibility and tensile strength. They shall have high heat and water resistance and vulcanize readily to natural rubber and to other synthetic rubber compounds used in the cable sheath.

3.1.1.2 Jacket. Synthetic rubber compounds used in the jacket with be polychloroprene or chlorosulphonated polyethylene. They shall be tough

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and resilient and shall have high flexibility, tensile strength and resistance to ozone attack, sunlight, heat, water, oil and abrasion. They shall vulcanize readily to other synthetic rubbers used in the insulation and to natural rubber.

3.1.1.3 Insulation - Jacket material combinations. Where insulation material is SBR, jacket material may be polychloroprene or chlorosulphonated polyethlene. Where insulation material is chlorosulphonated polyethylene or a combination of chlorosulphonated polyethylene and SBR, jacket material shall be chlorosulphonated polyethylene,

3.1.2 Reinforcing fabric or braid. Fabric or braid used to reinforce the synthetic rubber in the cable insulation shall be strong, flexible material made of natural or synthetic fibers. It shall be bonded firmly to the synthetic rubber materials and shall not be damaged by the temperature used to vulcanize the rubber. Reinforcing fabric or braid shall not be used in the cable jacket.

3.2 General requirements, The cable assembly shall be as specified hereinafter. In the event of any conflict between the requirements of this specification and the referenced drawings, the requirements of this specification shall govern,

3.2.1 Splices. No factory splices shall be allowed in individual sections of any cable unless approved by the command or agency concerned.

3.2.2 Flexural endurance. The cable assembly shall have high flexural endurance and shall be sufficiently flexible at all temperatures between minus 37°C and plus 60°C to permit ready handling and reeling on a drum 42 inches in diameter. "S" cable shall be suitable for storage on a reel 48 inches in diameter. Electrodes shall be suitablle for storage on a reel 34 inches in diameter.

3.2.3 Cable sheath. The cable sheath shall consist of an inner insulation in accordance with 3.1.1.1 and an outer jacket in accordance with 3.1.1.2, The insulation and jacket shall be vulcanized together in such a manner as to produce a single homogeneous sheath about the conductor and shall withstand the ply separation test specified in 4.4.5.3.

3.2.4 The sheath of flexible copper terminal-cables used on the “S” cable shall be the same as that used on the buoyant section of the “S” cab le. The molded sheath over terminal lug barrels shall be the same material as is used for the cable jacket.

3.2.4.1 The insulation of the terminal lugs shall be securely vulcanized to the sheath of the terminal cables and firmly bonded to the terminal barrels and flanges. When tested in accordance with 4.4.6 the adhesion shall be not less than 10 pounds per inch of width.

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3.2.5 Sheath materials used in pigtail cable shall be the same as that used in the flexible copper terminal-cables of 3.2.4 and 3.2.4.1.

3.2.6 Insulated sleeves. Material of insulated sleeves shall be polychloroprene or chlorosulphonated polyethylene.

3.2.7 Government furnished equipment. When specified (see 6.2), the Government will furnish the reels specified in 5.2.1.

3.3 Electrode cable. The electrode cable shall be a single conductor, self-buoyant, noninsulated cable for conducting electrical currents into seawater.

3.3.1 Construction. The electrode shall be constructed as shown on Drawing 30003-506AS1.00 In the event of any conflict between the requirements of this specification and the reference drawings, the requirements of this specification shall govern,

3.3.2 Core. Core of electrode shall consist of a 5/8-inch dacron polyester rope with polyethylene jacket as a stress member surrounded by cellular polyethylene as flotation material. Dacron polyester rope shall be made of prestretched yarns for low elongation and have a nominal breaking strength of 13,750 Ibs. Cellular polyethylene shall be foamed entruded over the dacron polyester rope to an average diameter of 2.40 inches ±0.10 inch. Specific gravity of core to be 0.57 + 0.05. Cellular polyethylene to be colored brown for sunlight protection. Overall specific gravity of complete electrode to be less than 1,00, Dacron polyester rope in core shall be well centered; core shall be as nearly round and uniform along length of core as possible.

3.3.3 Wire Wire shall be No. 6 AWG solik, continuous length, soft bare EC grade aluminum. Wires shall be cabled around core with and even tension and a left-hand lay at a pitch angle of approximately 20 degrees. The wire shall be applied so that it will remain in proper position and permit maximum flexibility of the finished electrode. The number of wires shall be 46 ±2 and shall be such as to insure a minimum coverage of 96 percent as determined by:

$$\text{percent coverage} = \frac{0.162 \text{ Sec } o \times 100N}{(D+0,162)ll}$$

where N = number of wires

o = pitch angle

D = average diameter of core, inches.

3.3.4 Serving. Serving shall consist of a continuous lenghth of No. 10 AWG EC grade H-19 aluminum wire in accordance with ASTM B230-71 wound in tight left-hand lay around conductors as shown in Drawing 30003-506AS100.

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3.3.5 Lug. One end of electrode shall be terminated with an aluminum stop. as shown on Drawing 30003-506AS100, Find #3, The stop shall be potted onto dacron polyester rope prior to installation of lug Find #5 with a glass fiber-epoxy resin compound consisting of approximately 75 percent epoxy resin and 25 percent glass fibers (percentages given by weight). Epoxy resin shall be equivalent to that manufactured by Kristel Kraft Inc., Palmetto, Florida, identified as "USNMDL Spec No, 3." Glass fibers shall be equivalent to that manufactured by Ren Plastics, Inc., Lansing 9, Michigan, identified as #701 1/32 inch milled fibers. After pouring potting compound, allow at least 24 hours curing time; remove excess compound before proceeding with assembly. Complete assembly by installing Serving Find #9, lug Find #5 and molding compound Find #7 as shown on Drawing No. 30003-506AS100,

3.3.6 Towing eye. Towing eye and installation are shown on Drawing 30003-506AS100.

3.3.7 Breaking strength. The minimum breaking strength of finished electrode shall be 6000 pounds.

3.4 Type "S" Cable. The type "S" cable shall be a single conductor, self-buoyant, insulated cable suitable for conducting d.c. electricity from a power source to an electrode.

3.4.1. Buoyancy. type "S" cable shall have a measured closed end specific gravity of not more than 0,94 and an open end specific gravity of not more than 0.96.

3.4.2 Construction. Type "S" cable shall be constructed as shown on Drawing 30003-506AS101.

3.4.3 Sheath thickness, The insulation and jacketing materials used in the sheath of the "S" cable shall have nominal thicknesses of 120 and 50 roils, respectively.

3.4.4 Terminals, Each end of the type "S" cable shall be terminated with a flexible connection consisting of (a) pares 9, 8, and 7 as shown on Drawing 30003-506AS101; (b) 6 feet, 0 inch of 1000 MCM 61 by 7/.0484 inch MHD copper cable; and (c) part 6 as shown on Drawing 30003-506AS101.

3.4.5 Physical and electrical properties. Type "S" cable shall have the following physical and electrical properties.

(a) Current carrying capacity - 2200 amperes continuous with the buoyant section in seawater at a temperature of 85 degrees F.

(b) Maximum d.c. resistance at 25 degrees C, - total resistance of cable shall. not exceed 0.008 OHM.

(c) Minimum breaking strength with terminals - 15,000 pounds,

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(d) D. c. contact resistance between parts 7 and 8 as shown on Drawing 30003-506AS101 shall be not greater than 4 microhms.

3.5 Pigtail cable. The pigtail cable shall be a single conductor nonbuoyant, insulated cable suitable for conducting direct current from a power source to an electrode.

3.5.1 Construction. Pigtail cable shall be constructed as shown on Drawing 30003-506AS101.

3.5.2 Sheath thickness. The Sheath of pigtail cable shall have a nominal thickness of 0.200 inch.

3.5.3 Terminals. Each end of the pigtail cable shall be terminated with part 6 as shown on Drawing 30003-506AS101.

3.5.4 Physical and electrical properties. Pigtail cable shall have the following physical and electrical properties.

(a) Current carrying capacity - 2200 amperes continuous with cable in seawater at a temperature Of 85 degrees F. .

(b) Minimum breaking strength with terminals - 15,000 pounds.

3.6 Insulating sleeve. The insulating sleeve shall be a synthetic rubber sleeve for protecting the terminals of the pigtail and "S" cable.

3.6.1 Construction. Insulating sleeves shall be constructed as shown on Drawing 30003-506AS101.

3.7 Identification.

3.7.1 Cable. The information listed in 3.7.1.1 shall be clearly molded into the insulation over the terminal barrels on both ends of 'IS" type cables.

3.7.1.1 Identification information. Identification information shall be as follows:

(a) Cable type designation ("S" cable only)

(b) Manufacturer.

(c) Year of manufacture.

(d) Serial number.

(e) Federal stock number.

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3.7.1.2 Serial numbers. Serial numbers will furnish by the command or agency concerned.

3.7.2 Reels.

3.7.2.1 Identification plates. Both flanges of each reel shall bave securely affixed thereto identification plates made by heat sealing between two approximately 0.020-inch thick clear, transparent, vinyl sheets,o accordance with L-P-535, a paper printed with the marking information required (see 5.4).Identification plates of other types providing equivalent legibility and durability may be used when specifically approved by the command or agency concerned.

3.7.2.2 Color series. Each reel of cable shall be identified with

fication shall consist of a stripe approximately 2 inches wide and colored as specified herein for the year of manufacture. The cycle of colors sall be repeated every fifth year.

Year of Manufacture		Identifying Color
1970	1975	White
1971	1976	Red
1972	1977	Green
1973	1978	Orange
1974	1979	Blue

3.7.2.2.1 The colored stripe shall be applied circumferentially over the lagging (see 5.2.4) and midway between the flanges, The stripe shall consist of one coat, commercial quality, outside paint of appropriate color. In addition to the stripe, both flanges of each reel shall be stenciled with 4-inch high figures to show the year of manufacture.

3.8 Cable sheath thickness. The minimum and maximum value of cable sheath thickness with reference to the nominal value shall be as follows:

(a) Average thickness of sheath shall be not less than 90 percent of the nominal value.

(b) Average thickness of sheath shall be not more than 125 percent of the nominal value, Buoyancy requirement shall not be changed.

(c) Any individual thickness shall be not less than 75 percent of the nominal value.

(d) The inside diameters of vulcanizing molds for cable terminations shall be the same asthe diameter of the flanges of the respective termination fittings. This requirement is to permit the proper installation of waterproofing boots and sleeves.

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3.9 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.1) an inspection lot shall consist not more than 25 cables of one type produced under essentially the same conditions, from the same raw materials, and offered for acceptance at one time.

4.2.2 One cable of each type shall be selected at random from each lot, and test samples specified in 4.2.2.1 or 4,2.2.2 shall be taken. The cable shall then be reterminated. Removal of the test sample shall not reduce the length of the cable to less than 148 feet (electrode) and 445 feet ("S").

4.2.2.1 Electrode.

4.2.2.1.1 A one-foot sample shall be cut from the towing eye end of electrode for use in performance of the tests specified in 4.4.7. The regerminated electrode shall be used for tests specified in 4.4.8.

4.2.2.1.2 In the event of failure of either sample or regerminated electrode to pass the tests specified in 4,4.7 or 4.4.8, the lot shall be rejected.

4.2.2.2 "S" Cables.

4.2.2.2.1 Five feet of buoyant conductor, along with the associated flexible terminal cable, connectors and lug, shall be cut from one end of "S" cable selected in accordance with 4.2.2. Two feet of buoyant conductor shall be cut from the end of the 5-foot sample and used in performance of the tests specified in 4.4.2 and 4.4.4. The remainder of the sample shall be terminated on the cut end in the same manner as the original cable and subjected to the tensile strength test specified in 4.~0..3.l. After completion of the test,one terminal lug shall be removed and used for the insulation adhesion test specified in 4.4.6.

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4.2.2.2.2. In the event of failure of any of the samples to pass the tests specified in 4.4.2, 4.4.3, or 4,4.4, the lot shall be rejected.

4.2.3 Circumferential specimens of cable sheath 2 inches wide shall be taken from each end of each "S" cable for use in accomplishing the tests specified in 4.4.5. Samples shall be removed before application of the terminal assemblies. Each sample shall be tagged with the date and serial number of the cable from which it was removed,

4.3 Examination. Production cable shall be examined to determine conformance with this specification as to dimensions, materials, workmanship and general construction. Total resistance measurements of "S" type cable shall be measured carefully, using accepted standard procedures.

4.4 Tests. The following tests shall be conducted:

4.4.1 Electrical.

4.4.1.1 Dielectric test. A dielectric test on each completed cable (except electrode) shall be performed by immersing the cable in water for 4 hours, after which time it shall be subjected to a potential of 2,500 volts r.m.s. (60 cycles per second) for 5 consecutive minutes between the conductor and ground. If the test should be interrupted, the total duration shall be increased 20 percent of the specified duration for each interruption. The cable shall withstand without failure the dielectric test. Any cable which fails to meet this requirement shall be rejected.

4.4.1.2 Insulation resistance. At completion of the dielectric test specified in 4.4.1.1 and while the cable is still in the water, the insulation resistance shall be measured using a 500-volt megohnmeter. Insulation resistance corrected to a temperature of 60°F shall be not less than 100 megohms as measured between the installed terminals and ground.

4.4.1.3 Contact resistance. A d.c. contact resistance test shall be made on each pressed connection of each complete cable to determine conformance with the resistance values specified in 3.4.5 (4 microhms).

4.4.2 Buoyancy. - "S" Type cable.

4.4.2.1 Closed end specific gravity. The 2-foot sample of "S" type cable shall be selected from each lot and shall be sealed by taping a thin rubber membrane over each end so as to exclude water. Each sample shall then be floated in a tank of fresh water at room temperature and lead weights applied along the length until the whole sample just sinks. The closed end specific gravity is then calculated from:

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$$\text{Specific gravity} = \frac{\text{Wt of cable in air}}{(\text{Wt of cable in air}) + (\text{Wt of lead in air}} \; 1004$$

4.4.2.2 Open end specific gravity. The same sample shall then have the rubber caps removed and shall be immersed in the tank of fresh water at room temperature with one end of the cable approximately at the water surface. The cable shall be held in this position for 5 minutes and the specific gravity determined as specified in 4.4.2.1, except without end covers on the cable.

4.4.2.3 The measured closed end specific gravity of the buoyant cables shall not be more than 0.94. The measured open end specific gravity shall not be more than 0.96.

4.4.3 Tensile Strength - "S" Type Cable.

4.4.3.1 The sample of "S" cable (see 4.2.2.2) shall have a tensile load of 15,000 pounds applied between end fittings for a 4-hour period. There shall be no permanent damage or deformation of the cable after test.

4.4.4 Compression test. A sample of "S" type of buoyant cable shall be selected from each lot and shall be subjected to a compressive force of 3,000 pounds for a period of 4 hours, applied between two flat plates 6 inches in length. The test shall result in no permnent deformationor damage to the cable.

4.4.5 Sheath tests.

4.4.5.1 Thickness. At least five separate measurements shall be made on each of the sheath samples taken in accordance with 4.2.3. The average of all the measurements taken on all samples from each cable shall be considered the average thickness. Where the materials used in the insulation and jacket differ, the requirements of this paragraph shall apply to the insulation and jacket individually. Tolerances specified in 3.8 shall apply.

4,4.5.2 Tensile strength. Five sheath samples from each lot shall be tested for tensile strength in accordance with method 4111 of Fed. Std. No. 601. The averaage tensile strength of the specimens shall be considered the tensile strength of the sheath and shall be not less than 900 p.s.i.

4.4.5.3 Ply separation test. On five sheath samples from each lot the bond between adjacent plys of taped construction sheath, between successive layers of extruded sheath and between the insulation and the jacketing compounds shall be tested using method 4221 of Fed. Std. No 601, except that the force shall be applied in such a manner as to tend to separate the individual layers of insulation and jacket along their mating faces. Bond between individual plys, the jacket and insulation, or between either of these and an intervening reinforcing tape or braid shall be not less than 10 pounds per inch of width at any point.

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4.4.5.4 Elongation test. Elongation tests in accordance with method 4121 of Standard Fed. Std. No. 601 shall be conducted on five jacket samples from each lot. The jacket samples shall be obtained by slicing the jacket material from sheath samples. Ultimate elongation shall be not less than 250 percent.

4.4.5.5 Oxygen, and sunlight resistance. The jacket material from a minimum of five sheath samples from each lot shall undergo an accelerated aging procedure consisting of the exposures outlined in methods 7111 and 7311 of Fed Std. No. 601. Twenty-four hours after acclerated aging is complete the tensile strength and elongation shall be not less than 60 percent of the values obtained in 4,4.5.2 and 4.4.5.4. Surface checks and cracks shall not be visible to the naked eye. As an alternative to test method 7311 for sunlight resistance a 100-hour exposure in type D test equipment per Standard ASTM-G-23-69 may be used employing a cycle of approximately 100 minutes of arc light exposure followed by approximately 18 minutes of exposure to arc light and water spray.

4.4.6 Terminal lug insulation adhesion test. A test specimen shall be prepared from the terminal lug selected in accordance with 4.2.2.2.1 and the adhesion of the insulation to the lug barrel tested by method 8031 of Fed. Std. 601. The sheath shall be reduced to a thickness of approximately 3/16 inch over the undistorted area of the lug during preparation of the specimen. Adhesion shall be not less than 10 pounds per inch of width.

4.4.7 Core sample. The diameter of the core of hte one-foot sample selected in accordance with 4.2.2.1.1 shall be measured as follows: At the approximate mid-point of the sample measure the diameter with a vernier caliper at three points approximately 120° apart. The average of the three measurements taken shall be considered the average diameter of the core. The average diameter shall be 2.40″ ± 0.10 Specific gravity of sample shall be measured. The specific gravity shall be 0,57 ± .05.

4.4.8 Regerminated electrode.

4.4.8.1 coverage. The number of wires, the pitch angle and the aver, age diameter as determined in 4.4.7 shall be used to determine the percent coverage as defined in Section 3.3.3. The percent coverage shall be not less than 96 percent.

4.4.8.2 Buoyancy, The electrode shall be placed in a tank of fresh water. The specific gravity of electrode will be considered less thatn 1.00 if electrode is floating after 15 minutes. . (see 3.3.2).

4.4.8.3 Tensile strength. The reterminated electrode shall have a tensile load of 5,000 pounds applied between lugs and towing eye for 15 minutes. During test, lug shall be restrained from rotating. There shall be no permanent damage to the electrode after test.

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5. PREPARATION FOR DELIVERY

5.1 One level of packing and packing, as specified in 5.2 and 5.3, shall apply for all shipping destinations and all storage periods for cable covered by this specification.

5.2 Packaging.

5.2.1 Type "S" cable shall be wound on reusable steel reels manufactured in accordance with Drawing s8101-1535866. Two "S" cables shall be wound on each reel. Both ends of the cable shall be securely fastened to the reel with rot resistant rope.

5.2.2 Electrode cables shall be wound on sturdy wooden shipping reels. Two electrode cables shall be wound on each reel. Reels shall be of suitable design, strongly constructed with flanges of sufficient diameter to protect the electrodes from injury in handling and rolling. Drum diameter of reels shall be not less than 34 inches,

5.2.3 Lagging. The cable wound on reels shall be covered with flexible waterproofed wrapping paper conforming to Specification uU-P-271 and secured to prevent unwrapping during handling and shipment. Reels shall be coxnpletely enclosed with wooden lagging with boards meeting each other. Lagging shall be 2-inch commercial lumber and shall be strapped with at least three l-inch wide steel straps, except that reels containing electrodes need have only two straps. Strap shall be stapled at intervals in accordance with the contractor's commercial practice. A minimum clearance of one cable diameter shall be provided between the inner face of the lagging and the waterproof wrapping material. When wooden reels are used, the lagging shall be cut even with the outside of the reel flanges and nailed with cement coated steel nails. Lagging of steel reel shall be in accordance with the. applicable steel reel drawing.

5.3 Packing. Cable packaged as specified in 5.2 shall not require overpacking. The method of packing or loading of reels shall be accomplished in a manner which will insure acceptance by carrier and will afford protection against physical or mechanical damage during shipment from the supply source to the using activity. The method of loading shall confrm to the American Association Gf Railroads Rules Governing the Loading Of Commodities and the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or order, each reel shall be marked as follows:

(a) Cable type
(b)Cable serial number
(c)Contract number
(d)Manufacturer's name and destination
(e)Federal stock number

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5.4.1 The information specified in 5.4 shall be provided on identification plates (see 3.7.2.1).

5.5 Pigtail cable and insulating sleeve shall be packed in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Ordering data. Procurement documenst should specify the folloowing.

(a) Title, number and date of this specification.
(b) Type of cable or cable assy required (see 1.2, 1.2.1, 1.2.2).
(c) Whether Government will furnish reels specified in 5.2.1.
(d) Lot quantity if special (see 4.2.1).
(e) Marking requirements other than those required by 5.4.

6.2 Subcontracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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APPENDIX

CORE CYLINDER FOR MAGNETIC MINESWEEPING CABLE

10. SCOPE

10.1 This appendix covers hollow plywood cylinders for the buoyant section,

20. APPLICABLE DOCUMENTS

20.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDEDERAL

MMM-A-181 - Adhesive, Room-Temperature and Intermediate-Temperature Setting Resin (phenol, Resorcinol, and Melamine Base).

Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Dimensions and weight. - The dimensions and weight of the hollow plywood cylinders shall be as follows:

Cable Type	Outside Diameter Inches		Shell Thickness Inches	Head Thickness Inches	Max. Avg. Wt. 300-1000 Cylinders Grams
S	2.50 ± 0,025	2 ±0 .035	0.110 ± 0.015	0 l 375 ± 0.035	54

30.2 Construction. - Construction requirements shall be as follows:

30.2.1 General. - All cylinders shall be first class material and workmanship throughout.

30.2.2 Plywood cylinder heads. - Cylinder heads shall be 5-ply pressure bonded poplar, gum or birch, or a combination thereof.

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30.2.3 Plywood cylinder shells. - Cylinder shells shall be 3-ply pressure bonded gum or birch. The plies shall be helically wound with adjacent plies in opposite directions. Veneer stock shall be of aircraft grade.

30,2.4 Adhesive. - Bonding of shell plies and between shell and heads shall be with a type I all-resorcinol adhesive conforming to MMM-A-181 applied and cured in accordance with manufacturer's detailed instructions.

30.2.5 Construction details, -

30.2.5.1 Hollow plywood cylinders. - Outer surface of cylinder heads shall be flush with ends of shell. The entire outer surface of the cylinder shall be sanded so as to provide a smooth base for the finish. Outer surface of cylinder heads shall be perpendicular to the cylinder axis, within a maximum deviation of 2 degrees.

30,2.6 Finish. -

30.2.6.1 Hollow plywood cylinders. - The entire outer surface of each cylinder shall be coated with a finishing material as follows:

First coat - Tumbled or dipped clear phenolic varnish.
Second coat - Tumbled, clear phenolic varnish.
Third Coat - Sprayed phenolic varnish containing aluminum flake.

Additional color coding may be applied at the option of the cable manufacturer.

40. QUALITY ASSURANCE PROVISIONS

40.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except-as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

40.2 Tests, - The following tests shall be conducted.

40.2.1 Diameter check, - The outer diameter of each cylinder shall be checked by means of "Go" "NO GO" gages. These are to consist of two l/4-inch thick metal plates bored to the diameter listed below with the edges of the holes rounded.

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Cable type	Nom. cylinder size Inches		NO GO gage Inches
"S"	2.50	2.527 .002	2.473 ± ,002

Cylinders shall pass through the GO gauge and be unable to pass through the NO GO gauge to be accepted.

40.2.2 Construction check. The shell thickness and head thickness (plywood cylinders), length and construction detail shall be checked on ten cylinders taken at random from each lot of 1,000 cylinder to determine that the requirements of 30,1 have been met. If any cylinder fails to meet any of the requirements, the representative lot shall be refected.

40.2.3 Weight check (plywood cylinders). A number (between 300 and 1,000) of cylinders shall be taken at random from each lot of 100,000 plywood cylinders and weighed to determine that the maximum average weight requirement of 30.1 has been met.

40.2.4 Physical data (plywood cylinders). Fifteen cylinders shall be taken at random from the quantity of cylinders weighed in accordance with 40,2.3. These cylinders shall be used for the environmental resistance tests of 40.2,5. Prior to performing the environmental resistance tests, the following data shall be taken and recorded.

(a) Diameter - Maximum and minimum, using diameter tape.
(b) Length - Maximum and minimum.
(c) Weight.
(d) Shell thickness - Masimum and minimum, shell construction (see 30.2.3).
(e) Head thickness, head construction (see 30.2.2).
(f) Perpendicularity (see 30.2.5.1).

40.2,5 Environmental resistance tests.

40.2.5.1 Tap water immersion test. Three of the fifteen cylinders shall be immersed in tap water at room temperature for 100 hours and weighed (w). The cylinders shall then be subjected to 100 C dry heat until 24 hour weight loss does not exced 0.1 gram per cylinder and reweighed (W). Moisture absorotion shall be calculated by the following formula and shall not exceed 25 percent:

$$\%MA = \frac{100W_w - W_D}{\%}$$

40.2.5,2 Cure test. Twelve of the 15 cylinders shall be exposed to a temperature of 100"F and 100 percent relative humidity for 50 hours. Immediately following this exposure, they shall be sealed in a moist:re-proof container such as a rubber bag having a volume not more than 10 percent greater than that of the cylinder contained therein. They shall

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then be subjected to a temperature of 298°F and a pressure of 70 p.s.i. for a period of two hours. They shall then be air cooled for one hour at room temperature before opening the container. The difference in diameter at the center of the cylinder as measured by diameter tape before and after test shall not be more than 1.5 percent of the cylinder diameter before test, and none of the cylinders shall collapse.

40.2.5.3 Radial crush test. Six of the cylinders precured as specified in 40.2.5.2 shall be subjected to a crush deformation test perpendicular to the axis of the cylinder. The deformation at load below shall not exceed 5 percent of the nominal cylinder diameter, Load to be applied through two flat plates.

type	O.D. of cylinder Inches	Load in pounds
W	2.50	2000

40.2.5.4 Boil delamination test. Three of the cylinders precured as specified in 40.2.5.2 shall be placed in boiling water for one hour and after removal shall show no evidence of delamination.

40.2.5.5 Failure to pass environmental resistance tests. If any cylinder fails to pass any of these tests, the representative lot shall be rejected.

40.2.6 Test reports. All test information shall be furnished to the prime contractor by the supplier.

50. PREPARATION FOR DELIVERY

50.1 Preparation for delivery shall be as specified by the cable manufacturer.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81953(AS) CABLE ASSEMBLY, MAGNETIC SWEEP MARK 17 MOD 1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL AIR SYSTEMS COMMAND
WASHINGTON, D.C. 20361

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